

PRODUCTION AI ENGINEERING · R1 + R2

REAL VS SIMULATED

Real vs simulated: what actually ran

Which parts of the R1 + R2 POC exercised the real mechanism, which are faithful local substitutes, what was recorded and what was injected.

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2 min read · Published 2026-10-07 · Real vs simulated

Real

the actual mechanism was exercised.

- Separate worker processes; 4 real SIGKILLS per runtime; a fail-stop and resume (check REL-R5-C01)
- HTTP over TCP to the providers, real socket timeouts, ECONNREFUSED on a closed port (check REL-R2-C02)
- A SQLite workflow journal with intent and result records (check REL-R3-C01)
- Provider-side idempotency windows, request deadlines and status queries (check REL-R1-C03)
- 96 calls to two local models through Ollama (check REL-R9-C01)
- A real-model end-to-end run: qwen3:8b deciding in every scenario x runtime, 78 calls, taped and replayed (check REL-R11-C07)

Simulated

a faithful local substitute for an external system.

- The credit, ticket and message providers, the CRM and the knowledge base, each with its own ledger
- Provider time, advanced to expire idempotency keys (S18)
- A scripted model for the fault scenarios, so that the recovery layer is the only variable

Recorded

captured model traffic, reused for deterministic re-scoring.

- The real-model tape (recovery_poc/runs//model-slice/tape.jsonl) (check REL-R10-C01)

Injected

a failure introduced on purpose.

- 25 preregistered faults (experiments/scenarios.toml) (check REL-R1-C01)
- 8 mutants of the recovery layer; X1 is the negative control (check REL-R7-C01)

Architecture

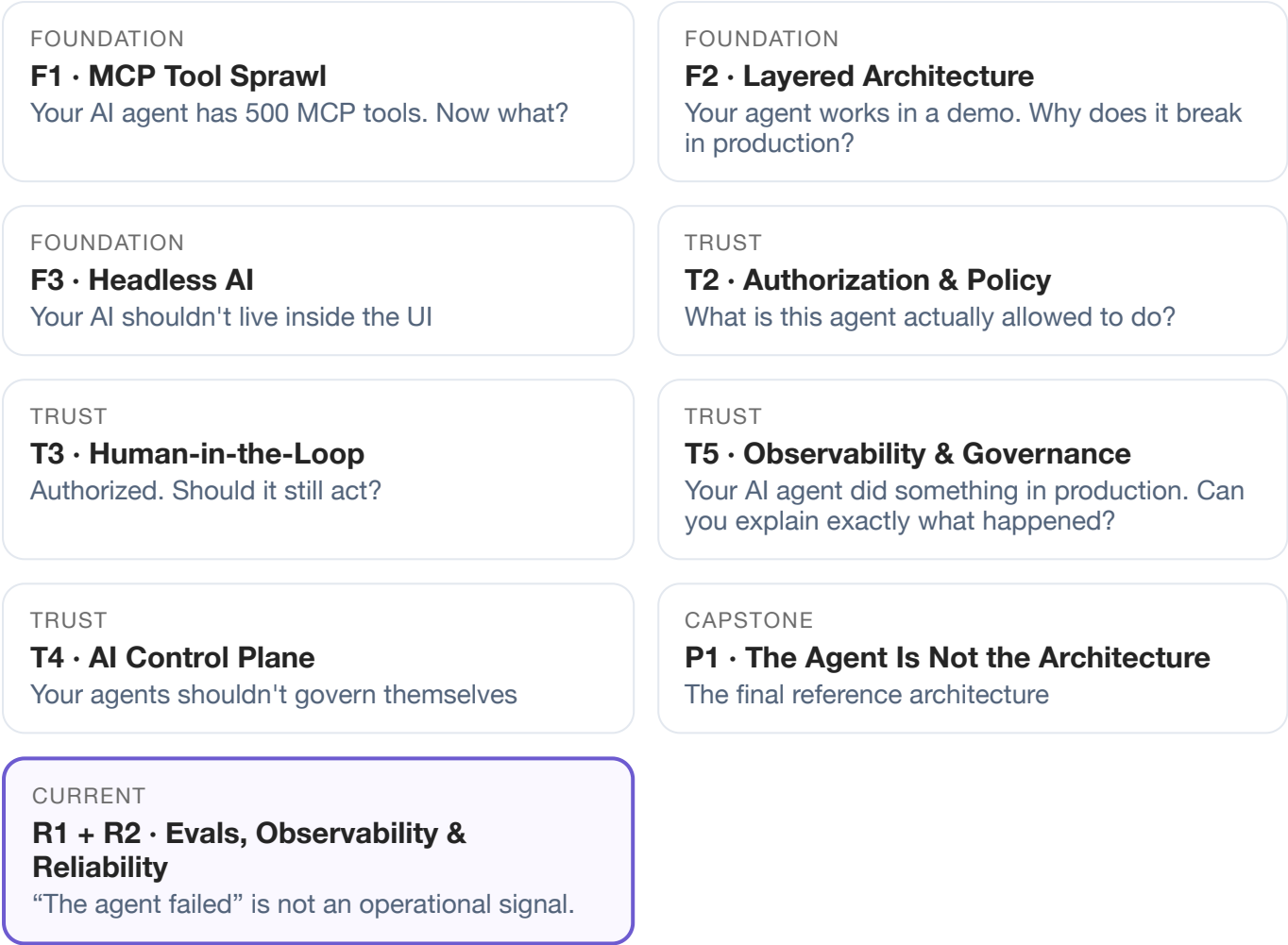
design this POC does not exercise.

- Load, backoff and retry budgets under contention; online evals on live traffic; a production escalation workflow
- Eventually consistent status queries; partial batch success; providers other than the four simulated

What the simulation idealises

- Reconciliation queries are immediately consistent once the request deadline has passed.
- Providers have no rate limits on status queries, no partial batch success and no clock skew with the runtime.
- Each scenario runs once; results are coverage of named faults, not a distribution of production failures.

PRODUCTION AI ENGINEERING · THE SERIES SO FAR



The full learning map (Start Here)

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Production AI Engineering · R1 + R2. Every measured number here is substituted at build time from `recovery_poc/runs/2026-10-07-recorded/facts.json`; the providers are simulated, the worker processes, crashes, sockets and timeouts are real, and the deterministic part of the run replays byte for byte. Figures carry their provenance.

- Medium edition Technical deep dive Evidence Check POC README Run summary
- Lab Console QA report